

SAMPLE LESSON · CASE STUDY: POTATO LATE BLIGHT

Can We Out-Breed Crop Disease?

A look inside a World Science Academy research session — where a real plant-disease problem becomes award-winning research.



TODAY'S CHALLENGE

A single disease once starved a nation.

Can we breed crops that resist it — and keep that resistance from breaking?



The Great Famine — Ireland, 1845–1852. Late blight destroyed the potato crop; over a million people died.

BY THE END, STUDENTS CAN

- ✓ Compare the tradeoffs of different disease-resistance strategies
- ✓ Explain why pathogens evolve past single-gene resistance
- ✓ Read real yield-and-cost data to make a recommendation

THE THREAT

Meet Late Blight

Caused by **Phytophthora infestans** — a fast-spreading, fungus-like water mould that can wipe out a whole field within days.

01 Spreads fast

Wind-blown spores jump field to field in cool, wet weather.

02 Devastating

Triggered the Irish Potato Famine (1845–1852).

03 A founding case

Its cause, proven in 1876, gave birth to plant pathology.



Blackened leaves & stems in the field



Brown rot spreading inside the tuber

Why a single defence fails

Variation

Mutations create new pathogen types.

Selection

A resistant crop kills off the susceptible ones.

Reproduction

Survivors multiply fast, spreading their genes.

Breakdown

The crop's single defence stops working.

The faster a pathogen varies and reproduces, the faster it evolves past a crop's defences.



KEY CONCEPTS

The language students build

Disease resistance

A plant variety's ability to limit a pathogen's growth and the damage it causes.

Pathogen

An organism that causes disease — here, the water mould *Phytophthora infestans*.

Natural selection

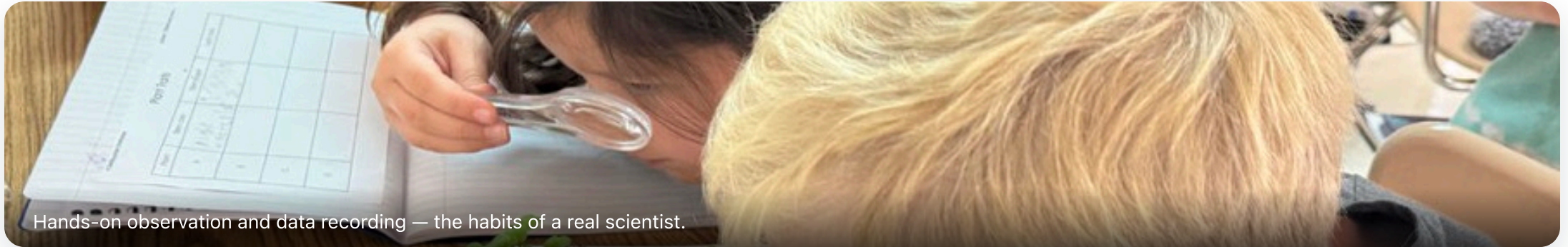
Individuals best adapted to their environment survive and reproduce.



The pathogen itself: *P. infestans* spores under the microscope.

HOW THIS SESSION RUNS

Students do, not just listen



Hands-on observation and data recording — the habits of a real scientist.

In-Person

- Real-time polling & word clouds
- Think-Pair-Share discussions
- Live dataset & graph analysis
- Case-study debates

Online

- Social annotation of case studies
- Micro-module self-checks
- Small-group discussion pods
- Scenario-based simulations

Which resistance strategy holds up?

Students compare four ways to defend a potato crop — and the trade-offs behind each.

Strategy	Yield loss (light)	Yield loss (outbreak)	Durability
Single R-gene	3%	52%	Low
Stacked R-genes	4%	8%	High
Variety mixture	14%	16%	Moderate
Susceptible + fungicide	6%	28%	High*

*Depends on continued spraying and no fungicide resistance. Illustrative teaching figures.

THE BIG IDEA

There is no single fix — only trade-offs

A single strong gene looks perfect until the pathogen evolves. Durable protection comes from **stacking defences** and **diversity** — the same logic behind vaccines, antibiotics, and crop breeding alike.

Evolution never stops

Diversity buys durability

Every choice has a cost

Crop vs. pathogen, through history

- 1845 **Irish Potato Famine.** Late blight destroys the crop; over a million die.
- 1876 **de Bary** proves a microbe — *P. infestans* — causes it. Plant pathology is born.
- 1900s Breeders move **R-genes** from wild potato into crop varieties.
- 1950s New pathogen races **overcome single R-genes** — resistance breaks down.
- Today** **Gene stacking & diversity** aim for durable, longer-lasting resistance.



Turning this into a science-fair study

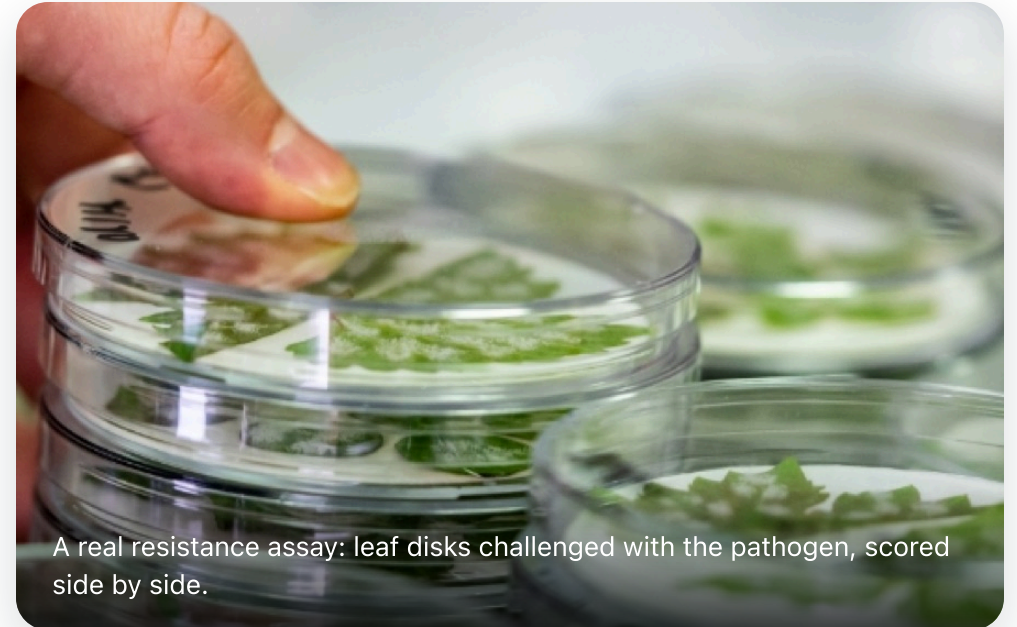
Ask a testable question

- Do mixed varieties slow disease vs. one variety?
- Which leaf traits predict resistance?
- How does temperature change spread rate?

Design a fair test

- Controls, replicates & a clear variable
- Measure & score disease objectively
- Graph the data — then argue from it

Our role: we coach students through each step — framing the question, running the experiment safely, analysing results, and presenting like a real researcher.



A real resistance assay: leaf disks challenged with the pathogen, scored side by side.

WHAT STUDENTS WALK AWAY WITH

Deliverables

01

A research question

A focused, testable hypothesis they can defend.

02

Experiment & data

A clean protocol, real measurements, and graphs.

03

Poster & talk

Competition-ready board and a confident presentation.

Scientific writing

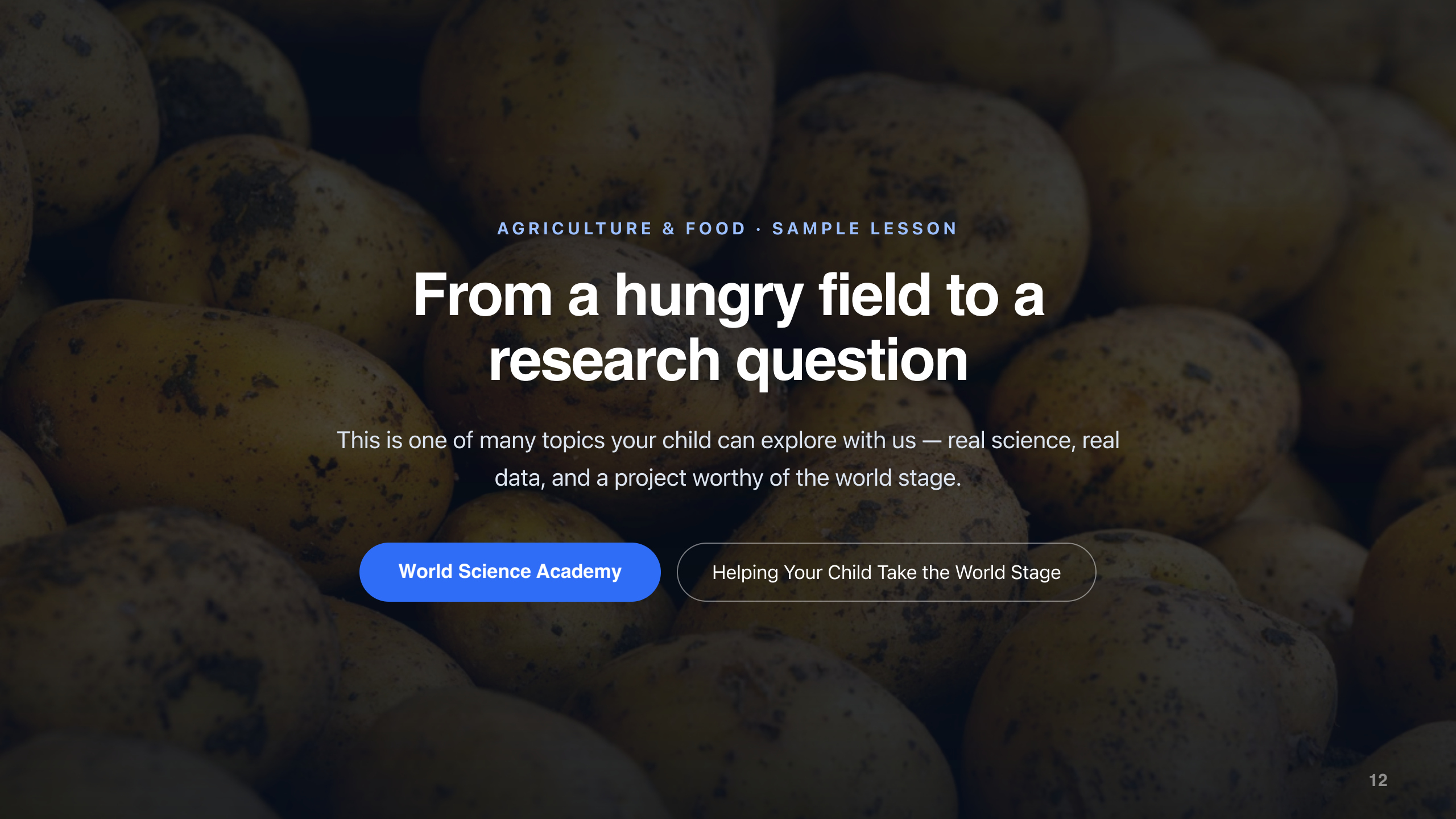
An abstract and report in real academic structure.

Judge-ready Q&A

Practice fielding tough questions like a researcher.



Presenting findings with confidence.

A close-up, slightly blurred photograph of several potatoes, showing their characteristic brown skin with some darker spots. The lighting is soft, creating a warm, earthy tone. The potatoes are piled together, filling the entire frame.

AGRICULTURE & FOOD · SAMPLE LESSON

From a hungry field to a research question

This is one of many topics your child can explore with us — real science, real data, and a project worthy of the world stage.

World Science Academy

Helping Your Child Take the World Stage